MUHAMMAD FAIZAN YOUSAF

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Professional Summary

Computer Science student with experience in full-stack web development using React, Next.js, Node.js and Django. Built AI-assisted features using NLP, Generative AI, and Retrieval-Augmented Generation (RAG). Comfortable working with REST and FastAPI services and SQL/NoSQL databases. Interested in building practical applications that combine AI with modern web systems.

Education

National University of Computer and Emerging sciences *August 2022 - June 2026*

BS Computer Science
Relevant Courses: Artificial Intelligence, Web Programming, Database Systems, NLP, Object-Oriented Programming, Data Structures and Algorithms
CGPA: 3.30/4.0

Technical Skills

- **Programming Languages:** C, C++, Python, JavaScript, C#
- **Frontend Development:** React, Next.js
- **Backend Development:** Node.js, Express.js, Django, FastAPI, Firebase, RESTful APIs
- **Database Management:** MongoDB, MySQL, Firebase Firestore
- **AI and Machine Learning:** NLP, Generative AI, Supervised & Unsupervised Learning, Fine-tuning, FAISS, Feature Engineering
- **Tools & Technologies:** Git, GitHub, VS Code, Linux, Postman, Docker (basic)

Professional Experience

Intern - Digital Genei (Onsite) June 2025 - July 2025
- Developed a speech-to-text pipeline using Whisper for meeting transcription and applied NLP summarization to extract key discussion points.
- Implemented semantic search using FAISS for efficient vector indexing and similarity-based retrieval.

Intern - Skillify Zone (Remote) July 2025 - August 2025
- Built a loan approval prediction system using Pandas and Scikit-learn with structured dataset preprocessing and feature engineering.
- Performed data cleaning, exploratory data analysis, and model evaluation to improve prediction accuracy.

Projects

HealthMate (Final Year Project)
Flutter, Dart, Firebase, Firestore
- Developed a cross-platform mobile health application with Flutter and Dart that enables users to securely submit lifestyle data, health metrics, and genetic SNP questionnaire responses.
- Implemented secure authentication, real-time data storage, and wearable device integration using Firebase services to enable continuous health monitoring and structured data collection.

AI Knowledge OS (AI-Powered Knowledge & Project Management Platform)
Next.js, React, Node.js, Express.js, MongoDB, RAG, NLP, Vector Embeddings
- Built a full-stack application for managing project notes and documentation with semantic search using RAG and vector embeddings.
- Implemented JWT-based authentication and an AI chat interface that retrieves and summarizes relevant project information.

RAG for diagnostic reasoning for Clinical Notes
Natural language processing, Prompt engineering, Python, Vector database
- Implemented a Retrieval-Augmented Generation (RAG) based clinical Q&A system using MIMIC-IV healthcare dataset, integrating LLMs with a Streamlit interface to enable intelligent medical information exploration.
- Designed a context-aware medical assistant that retrieves relevant clinical records in real time using vector embeddings and semantic similarity search to generate context-aware responses.

VerbatimAI (AI-Powered Meeting Intelligence System)
Python, Machine Learning, NLP, Streamlit, Vector Embeddings
- Developed a meeting analysis system that converts recorded meetings into structured insights using speech-to-text transcription and NLP-based summarization.
- Designed an interactive Streamlit interface enabling semantic search and insight discovery from meeting transcripts using embedding-based similarity retrieval.

Certifications

Coursera – Machine Learning Specialization
(Supervised Learning, Unsupervised Learning, Advanced Learning Algorithms)